ELI — Finetuning Guide

ELI v2.5.21

October 2026

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ELI Fine-Tuning Guide — Train a Model the Correct Way (End-to-End)

A meticulous, do-this-exactly walkthrough for turning **a base model you choose** into an **ELI-native GGUF**, then loading it back into ELI. Covers the *why* as well as the *how*, so it's a guide you can reason from, not cargo-cult.

Method: QLoRA (4-bit base + LoRA adapters). **Pipeline:** the scripts in `training/` (extract → train → merge → convert → install). **Default base:** `Qwen/Qwen3-8B` — but the whole guide is written around a single `BASE=...` you set once (Stage 0), so the *same commands* train **any** model you pick into a GGUF.

> Scope: this is **fine-tuning** (adapting a pretrained model), **not** pretraining from scratch (that needs thousands of GPU-hours and is out of reach here — and unnecessary).
>
> Hardware: QLoRA keeps this on a single consumer GPU. The defaults target an ~8 GB card; the `--gpu-mem` / base-size knobs in Stage 2 scale it up or down for whatever you have. No specific GPU is assumed anywhere.

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0. The five principles (read before touching anything)

1. **Persona stays dynamic — never bake it into the weights.** ELI's persona is live (`persona_updater` + overlay + the per-turn memory brief). You fine-tune the **stable layer only**: ELI's *voice / manner* and its *behavioral contracts* (No-Fake-Actions, grounding, dry register). You do **not** train in user facts, memories, dates, or a fixed persona prompt. The old `models/extract_training_data*.py`

dumped `memories[-50:]` in as system prompts — that froze state. The new `extract_eli_dataset.py` deliberately does not.

2. **LoRA, not full fine-tune.** A full fine-tune of an 8B model needs ~60–100 GB VRAM. LoRA trains tiny low-rank adapters (<1% of params) on top of a frozen 4-bit base → fits 8 GB. For a *style/voice* shift (what we want), LoRA is not a compromise — it's the right tool.
3. **Data quality data quantity.** A few hundred clean, on-voice examples beat thousands of noisy ones. Garbage in → garbage (or overfit) out.
4. **Match the chat template to the target model.** Qwen3 uses ChatML. The dataset MUST be formatted with the *target model's* tokenizer template, or the model learns the wrong control tokens and produces garbage. (`extract_eli_dataset.py` does this via `tokenizer.apply_chat_template`.)
5. **The base model must have enough context for ELI's brief.** ELI's runtime prompt is ~7–9k tokens. A 4k-context base (phi-3-mini) **cannot** run ELI — it produced literal garbage (`_....,chknce/you,//`). Qwen3-8B's 40,960 context is why it's the pick. (This is enforced at load by the ctx tuner — see §7.)

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1. Concepts you must understand (so this is rigorous, not ritual)

- **Base vs Instruct weights:** you fine-tune **HF weights** (a directory of `.safetensors` + tokenizer), *not* a GGUF. GGUF is the **inference** format ELI loads; HF is the **training** format. They are not interchangeable — you'll download HF for training and produce a GGUF at the end.
- **LoRA (Low-Rank Adaptation):** freezes the base weights and learns two small matrices per target layer whose product is added to the original weight. `r` = the rank (capacity of the change); `alpha` = a scaling factor (effective LR multiplier ≈ `alpha/r`). Small `r` (8) = cheap, good for style; large `r` = more capacity, more overfit risk.
- **QLoRA (4-bit):** the frozen base is quantized to 4-bit (nf4) to fit VRAM; the LoRA adapters train in fp16. Quality loss from 4-bit base is negligible for fine-tuning.
- **`target_modules`:** which weight matrices get adapters. We adapt all attention + MLP projections (`q/k/v/o_proj`, `gate/up/down_proj`) — the standard full coverage.
- **Quantization (inference):** the merged model is converted to GGUF and quantized — `Q4_K_M` for an 8 GB card (best size/quality balance), `Q5_K_M`/`Q8_0` if you have headroom.
- **Catastrophic forgetting / overfitting:** too many steps or too high `r`/LR makes the model *memorize* your data and *forget* its general ability (and parrot you robotically). Defend with: low `r`, modest LR, **few steps**, and watching the loss (see §4).
- **`n_ctx_train`:** the base model's trained context window. It sets the ceiling for inference ctx; ELI reads it from GGUF metadata at load (the ctx tuner).

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2. Environment setup (once)

```
# Use the project venv (CUDA already verified there).
cd ~/Desktop/ELI_MKXI-main_MAY_NEWEST
.venv/bin/python -m pip install --upgrade unsloth trl transformers datasets accelerate bitsandbytes huggingf

# llama.cpp — needed to convert HF -> GGUF and quantize (if not already present):
#   git clone https://github.com/ggerganov/llama.cpp && cd llama.cpp && make
#   (note its path; you'll point merge/convert at it)
```

Sanity checks:

```
.venv/bin/python -c "import torch; print('CUDA:', torch.cuda.is_available(), torch.cuda.get_device_name(0))"
.venv/bin/python -c "import unsloth, trl, transformers, datasets; print('train stack OK')"
```

> **Hardware honesty:** QLoRA-training an 8B on 8 GB works but is **tight and slow** (heavy CPU offload). If it crawls or OOMs, drop to **Qwen3-4B** as the base (`--base-model Qwen/Qwen3-4B`) — trains far more comfortably, still 32k+ context, still a strong ELI substrate.

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3. Step 0 — Choose your model, then get its base HF weights

0a. Choose the base (the decision that matters most)

You train **HF weights**, not a GGUF (see §1). Pick a base whose context is **≥16k** (ELI's brief is ~7–9k tokens — a 4k base produces literal garbage). Set it **once** as a shell variable and every command below is identical regardless of which model you chose:

```
export BASE="Qwen/Qwen3-8B"          # default — strong ELI substrate, 40k ctx, fits ~8 GB at Q4
# export BASE="Qwen/Qwen3-4B"        # lighter — trains faster/comfier, still 32k+ ctx
# export BASE="microsoft/phi-4"      # 14B dense (MIT), needs more VRAM/offload
# export BASE="tiiuae/Falcon3-10B-Instruct"   # or any HF instruct model with ≥16k ctx
export NAME="eli-$(basename "$BASE" | tr '[:upper:]' '[:lower:]')"   # output gguf base name
```

Want	Pick	Why
Best all-round default	`Qwen/Qwen3-8B`	40k ctx, reasoning-capable, fits 8 GB at Q4
Lightest / fastest train	`Qwen/Qwen3-4B`	comfortable on 8 GB, still ≥32k ctx
Strong reasoning, more VRAM	`microsoft/phi-4`	14B dense, MIT-licensed
Try something else	any HF instruct repo	**must have `n_ctx_train` ≥ 16k**

> The GGUF keys in ELI's download catalog (`qwen3-8b`, `phi-4`, …) are *inference* builds. For **training** you need the matching **HF** repo (the `BASE` above), not the GGUF.

0b. Download the base HF weights (~16 GB for an 8B; the one online step)

```
.venv/bin/huggingface-cli download "$BASE" --local-dir "training/base/$(basename "$BASE")"
export BASE_LOCAL="training/base/$(basename "$BASE")"   # use this in the commands below
```

You can also pass the HF id directly (`--base-model "$BASE"`) to let HF cache it instead.

---

4. The pipeline (the meticulous part)

Step 1 — Extract the dataset (voice/manner only)

```
.venv/bin/python training/extract_eli_dataset.py \
    --base-model Qwen/Qwen3-8B \
    --out training/datasets/eli_voice.jsonl
```

What it does: reads your stored conversations (`artifacts/conversations/*.json`), keeps the USER↔ELI turns, formats each with **Qwen3's chat template**, and writes `{"text": ...}` JSONL. By default `--system-mode none` and **memory-state injection is excluded** (persona stays dynamic).

Curate before you train (this is where quality is won): - Open the JSONL. Remove turns where ELI was *wrong, off-voice, or fabricating* (you don't want to teach the failures we spent this week fixing). - Keep turns that exemplify the voice you want: dry, direct, honest, no fake actions, grounded. - Aim for **a few hundred good examples**, not thousands of mediocre ones. - Optional: hand-write a dozen "gold" exemplars of perfect ELI behavior (refusing to fake an action, admitting uncertainty, the dark-humor register) and append them — high-signal.

Step 2 — Train the LoRA adapter

Always do a smoke run first (10 steps) to confirm the loop before committing hours:

```
.venv/bin/python training/train_lora.py \
    --base-model Qwen/Qwen3-8B \
    --dataset training/datasets/eli_voice.jsonl \
    --out training/runs/eli-qwen3-8b-lora \
    --max-seq-len 4096 --max-steps 10
```

Then the real run (`--max-steps 200–400`). Hyperparameters, and how to set them:

Flag	Default	What it controls	How to tune
`--max-steps`	300	total optimizer steps	start 200–400; **watch loss** (below). More ≠ better.
`--lora-r`	8	adapter capacity	8 for voice; 16 only if voice isn't "sticking". Higher → overfit.
`--lora-alpha`	16	adapter scaling (≈2×r)	keep ≈ 2×r.
`--lr`	2e-4	learning rate	1e-4 (gentle) … 2e-4 (standard). Too high → loss spikes/garbage.
`--max-seq-len`	4096	max example length	raise to fit your longest multi-turn example; costs VRAM.
`--batch` / `--grad-accum`	1 / 8	effective batch = 8	leave at 1×8 on 8 GB; raise grad-accum for a smoother loss.
`--gpu-mem` / `--cpu-mem`	5.5GiB / 20GiB	VRAM/RAM split	lower `--gpu-mem` if OOM; the rest spills to CPU (slower).

Reading the loss (your only real gauge): - It should **decrease then flatten**. A gentle decline to a plateau ≈ healthy. - **Loss → near 0** = memorizing your data (overfit). Stop earlier / lower r/steps. - **Loss flat/noisy from the start** = LR too low, data too small, or template mismatch. - Save adapter checkpoints; if a later checkpoint sounds *more robotic*, use an earlier one.

Output: a LoRA adapter at `training/runs/eli-qwen3-8b-lora/`.

Step 3 — Merge → convert → quantize → install

```
.venv/bin/python training/merge_and_convert.py \
    --base-model Qwen/Qwen3-8B \
    --adapter training/runs/eli-qwen3-8b-lora \
    --out-name eli-qwen3-8b --quant q4_k_m
```

Produces and installs `models/eli-qwen3-8b-q4_k_m.gguf`. (Manual llama.cpp fallback if Unsloth's GGUF export fails — see `training/README.md`.)

Step 4 — Load it in ELI

Model menu / Auto Detect → pick the new GGUF. The **ctx tuner** reads its real `n_ctx_train` (40960), right-sizes ctx to ~16k, and keeps it on the GPU. The **dynamic persona + memory** are still supplied fresh by the runtime — your fine-tune just made the *substrate* speak ELI.

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5. Verification — is the fine-tune actually good?

Don't trust "the loss went down." Load it and check **four** things: 1. **Voice:** does it sound like ELI (dry, direct, honest) *without* the heavy persona brief leaning on it? Compare a few prompts vs base Qwen3-8B. 2. **Contracts hold:** ask it to "pause spotify" when it can't, or to analyze a file — it must **not fabricate** an action/result (No-Fake-Actions). Ask a grounded question — it must not confabulate. 3. **Persona still dynamic:** confirm the runtime overlay/memory still steer it (e.g., it picks up a new stated preference this session). If it ignores the live brief and only parrots training data → you overfit; retrain gentler. 4. **No capability regression:** a couple of reasoning/coding prompts should be ~as good as base Qwen3-8B. If markedly worse → overfit; fewer steps / lower `r`. Keep a tiny held-out set of prompts and run it after every training run — that's your eval.

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6. Troubleshooting (cause → fix)

Symptom	Cause	Fix
Garbage output (`_...,/you//`)	ctx > model's `n_ctx_train`, or wrong chat template	ensure base ≥16k ctx; confirm extract used the *target* tokenizer template
CUDA OOM during training	base too big for VRAM	lower `--gpu-mem`; smaller base (Qwen3-4B); lower `--max-seq-len`; `--batch 1`
Loss won't drop	LR too low / data too small / template mismatch	raise `--lr` to 2e-4; add data; re-check template
Model parrots you, ignores live persona	overfit	fewer `--max-steps`, lower `--lora-r`, earlier checkpoint
Worse at reasoning than base	overfit / too-high rank	lower `r`, fewer steps
GGUF convert fails	Unsloth export quirk	use the manual llama.cpp `convert_hf_to_gguf.py` + `llama-quantize` path (README)
Fine-tune fabricates actions	trained on ELI's *bad* turns	scrub the dataset of fabrication/off-voice turns; add gold no-fake exemplars

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7. How this ties into ELI's runtime (so the loop is closed)

- **Model-agnostic loader:** drop the produced GGUF in `models/`; ELI finds it, no config.
- **Ctx tuner (shipped):** reads the model's real `n_ctx_train` from metadata, right-sizes ctx, maximises GPU layers, and *warns* if a model's context is too small for ELI's brief — so a bad-fit base can never silently degrade you again.
- **Contracts stay in code:** No-Fake-Actions, grounding, fenced-JSON tolerance, empty-<think> recovery are **runtime guards** (this week's commits). The fine-tune *reinforces* the voice; the guards *enforce* the behavior. Belt and braces.

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8. Quick reference (copy-paste — works for ANY chosen base)

```
# choose your model ONCE (Stage 0)
export BASE="Qwen/Qwen3-8B"                                  # ← change this to train a different model
export NAME="eli-$(basename "$BASE" | tr '[:upper:]' '[:lower:]')"
export BASE_LOCAL="training/base/$(basename "$BASE")"
```

```
# one-time setup
.venv/bin/python -m pip install --upgrade unsloth trl transformers datasets accelerate bitsandbytes huggingf
.venv/bin/huggingface-cli download "$BASE" --local-dir "$BASE_LOCAL"

# pipeline (extract → train → merge/convert/install)
.venv/bin/python training/extract_eli_dataset.py --base-model "$BASE_LOCAL" --out training/datasets/eli_voi
.venv/bin/python training/train_lora.py          --base-model "$BASE_LOCAL" --dataset training/datasets/eli_
.venv/bin/python training/merge_and_convert.py   --base-model "$BASE_LOCAL" --adapter "training/runs/$NAME-
# then: load models/$NAME-q4_k_m.gguf in ELI (Model menu / Auto Detect)
```

File map

- `training/extract_eli_dataset.py` — dataset (voice only, persona-dynamic-safe)
- `training/train_lora.py` — QLoRA trainer (parameterised, Qwen3-8B default)
- `training/merge_and_convert.py` — merge → GGUF → quantize → install
- `training/README.md` — quick-start; this doc — the full rationale + correctness
- `training/datasets/`, `training/runs/`, `training/base/` — data, adapters, base weights (created locally when you run the pipeline)

Golden rules, one more time

1. Fine-tune **voice + contracts**, never the dynamic persona/memory.
2. **Clean** the dataset; quality over quantity.
3. **Smoke-run**, then watch the **loss**; stop before it memorizes.
4. Base model must have the **context** to hold ELI's brief (≥16k).
5. **Verify on real prompts** (voice + contracts + dynamic-persona + no-regression), not on loss.

Appendix — the guided path

Everything above remains the rigorous manual route, and it is still the one to read if you want to understand what is happening. The same pipeline is also driven from **Labs ▸ Training**, which is the recommended path on an unfamiliar machine because it *reports* rather than assumes.

Step 1 — Hardware. Accelerator and vendor (NVIDIA / AMD ROCm / Apple MPS / Intel XPU), free VRAM, whether `bitsandbytes` is present for 4-bit, which of `torch / transformers / peft / accelerate / datasets` are missing, and every trainable Hugging Face base directory it can find with its family and size. If the machine cannot train, it says so in a sentence instead of starting a job that never finishes.

Step 2 — Target. Declare a target against any local HF base. The family is read from the model's own `config.json`; you do not have to know it. Built-in targets are listed and cannot be deleted. This replaces editing `ALLOWED_TARGETS` in source.

Step 3 — Data. The review queue — the step that was previously impossible. Candidates are mined from your own stored conversations, triaged (obvious rejects removed, questionable rows flagged), and then approved, edited or rejected by you. "Approve all clean rows" handles the bulk; the flagged remainder is the part that actually needs judgement. Saving writes the reviewed, target-scoped rows the trainer demands.

Step 4 — Train. Pick a recipe:

Recipe	Steps	Seq len	Accum	LR	Rough time
Light	60	256	4	2e-4	minutes
Standard	300	512	8	1e-4	tens of minutes
Deep	1200	1024	16	5e-5	hours

Raw parameters stay available under **Advanced**. “Check readiness” runs the whole DAG as a dry-run and prints every blocker. “Start training” runs it for real with live step and loss, and a cancel that stops at a step boundary so the adapter directory is left sane.

What the wizard does NOT do

Merge the adapter into the base and convert to GGUF. That is still `training/merge_and_convert.py`, and it still needs `unsloth`, which is in no requirements file. Until you run it, the new adapter is not the model ELI is serving — and the wizard says exactly that when a run finishes rather than implying otherwise.

If it refuses

- *“dataset has no usable rows”* — nothing is approved yet. Step 3.
- *“base model is GGUF”* — a `.gguf` is an inference artifact. You need a Hugging Face model **directory**; see §3 above.
- *“base_family mismatch”* — the target says one family, the checkpoint on disk says another. Re-declare the target against the right folder.
- *“needs ~N GiB, you have M”* — free VRAM, shorten the sequence length, or install `bitsandbytes` for 4-bit.